Universal Governance Kernel (UGK)

Accountable Authority as an Execution Substrate

The Universal Governance Kernel (UGK) is a governance-first execution architecture.

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Abstract

The Universal Governance Kernel (UGK) is a governance-first execution architecture. Instead of asking only whether an actor has a permission or whether a mechanism can perform an operation, UGK asks what is intended, whose purpose is being served, which authority can bind the decision, which evidence is admissible, what disposition follows, and what exact effect and receipt result.

UGK separates Steward, Governor, and Executor as functions; requires a deployment to be founded under a declared charter before ordinary effects are admitted; types operations through a schema registry; and retains admission, refusal, execution, failure, and verification records as a reviewable governance history. Its central discipline is that decision authority must be explicit and execution must remain downstream of a committed decision.

The architecture can make authority claims and action lineage inspectable. It cannot make a charter legitimate, an authority deserved, evidence true, a purpose ethical, or a decision wise merely by representing and hashing them. This paper specifies the architecture and its failure tests. It does not claim that a conforming implementation, deployment, or cryptographic proof has been supplied.

Source, Evidence, and Standing

This is an AI-assisted integrated edition derived from one exact educational brief. Its custody and comparison manifest is `work/paper-saturation-v1/evidence/universal-governance-kernel-brief/integration-custody-manifest.v0.1.json`, SHA-256 `6B934D0CF3BB772E462EB2F7BCE329329E3E0978835FA156BD78A0772E18D2A4`. The source remains unchanged.

The kernel framing, governance chain, functional roles, charter, receipts, schema tiers, authority postures, authority-laundering analysis, and limits are author material. The formal event/state model, distinction grammar, invariants, corpus interfaces, full realization funnel, expanded failure tests, and editorial alternatives are AI-assisted contributions explicitly distinguished from the original author material.

No implementation or conformance evidence was inspected. The corpus contains several named UGK systems, profiles, codices, and implementation candidates. Their identity and version relationships remain unadjudicated; this brief does not merge or certify them.

1. The Governance Gap

Conventional execution paths often compress several questions into one permission check:

```
request -> permission -> action
```

That path can establish that a credential or role satisfied a rule. It may not establish whose purpose the action serves, which delegation is operative, which authority can bind the decision, what evidence was evaluated, what was refused, or how the effect can later be reconstructed and challenged.

UGK expands the path:

```
Intent -> Will -> Authority -> Decision -> Commitment -> Execution
-> Result Receipt
```

The objective is not universal prevention. It is accountable authority: significant effects should have a reconstructable legitimacy claim, decision basis, bounded effect, and evidentiary trail.

2. Distinction Grammar

Identity answers who or what is acting. Identity is not authority.

Capability answers what operations are technically reachable. Capability is not permission or legitimacy.

Intent describes the requested action and target.

Will names the represented principal, purpose, or mandate the request claims to serve. A declared purpose does not generate authority.

Authority is a bounded basis on which a decision may be made for a named action, target, jurisdiction, time, and delegation chain. Authority is not ambient and does not arise from system acceptance alone.

Evidence is the material admitted for evaluation under a declared rule and standing.

Decision is the Governor's typed disposition, such as admit, refuse, defer, conflict, or require amendment.

Commitment is the durable pre-effect record binding the admitted request, decision, authority, exact inputs, and permitted effect.

Execution is the attempted effect performed only within the commitment.

Receipt is evidence of a governance or execution event. A receipt's claim ceiling is bounded by what was observed and committed.

Legitimacy is a larger judgment about whether the charter, authority, process, and result deserve binding force. UGK can expose the inputs to that judgment but cannot manufacture it through structure alone.

3. Functional Separation

3.1 Steward

The Steward function proposes an intent and identifies the principal or purpose it claims to serve. It does not approve its own authority merely by making a request.

3.2 Governor

The Governor function resolves applicable policy and authority, evaluates evidence, and emits a typed decision with reasons and scope. It must remain distinguishable from the proposer and effecting mechanism even when one process hosts all three functions.

3.3 Executor

The Executor function attempts only the committed effect. It cannot widen target, action, scope, duration, or delegation. Technical capacity beyond the commitment remains unreachable through the admitted path.

These are constitutional functions, not necessarily separate people, services, or machines. A realization may combine them physically only if their logical boundaries and evidence remain inspectable.

4. Governed Event Model

Let a request be:

$$q = (i, w, a, x, e, p, t),$$

where i is intent, w represented will or purpose, a asserted authority, x exact target and operation, e evidence, p governing policy and charter state, and t evaluation time.

The Governor returns:

$$J(q) = (d, r, s, c),$$

where d is disposition, r reason, s authorized scope, and c the pre-effect commitment when and only when the request is admitted.

Execution returns:

$$X(c) = (o, v),$$

where o is the observed outcome and v the result evidence. The final receipt binds q, $J(q)$, c, o, and v without claiming more than those records establish.

An admission decision is not proof that the effect occurred. A result receipt is not retroactive proof that the decision was legitimate. A failure receipt can prove an attempted execution and observed failure without proving that no external side effect occurred unless the observation boundary supports that claim.

5. State and Transition Semantics

A deployment begins **unfounded**: ordinary application effects are not admitted. A founding transition binds a charter, identity roots, jurisdiction, policy versions, schema registry, amendment procedure, and founding provenance. Founding records a claimed authority regime; it does not prove the charter morally or legally legitimate.

A governed event moves through:

```
UNFOUNDED
  -> FOUNDED
  -> REQUESTED
  -> EVALUATED
  -> COMMITTED or REFUSED/DEFERRED/CONFLICTED
  -> EFFECT_ATTEMPTED
  -> EFFECT_CONFIRMED, EFFECT_FAILED, or EFFECT_UNKNOWN
  -> RECEIPTED
```

Refusal, deferral, conflict, and unknown are terminal for the current effect unless a new, separately evaluated request is created. They are not exceptional gaps to be omitted from history.

6. Charter, Amendment, and Authority

A charter should minimally declare:

- governed domain and jurisdiction;
- identity and trust roots;

- functional roles and separation requirements;
- authority types and delegation rules;
- operation schemas and effect classes;
- evidence and decision rules;
- refusal and unknown semantics;
- receipt and verification obligations;
- amendment, revocation, recovery, and appeal procedures;
- privacy, retention, and disclosure bounds.

Authority must be named, scoped, time-bound where applicable, and traceable through delegation. A credential can authenticate an actor without proving that the actor's authority applies to the requested effect. A charter can authorize a procedure without proving that the surrounding institution is just.

The source mentions trace-only, alternate-trace, and prevention authority. These are useful candidate postures, not yet an exhaustive authority ontology. A deployment must define exactly what each posture can observe, decide, block, simulate, or effect.

7. Schema Registry

The source proposes three operation tiers:

- **Tier 0:** kernel operations such as admission and refusal;
- **Tier 1:** universal governance operations such as session, evidence, audit, and verification events;
- **Tier 2:** application-defined operations constrained by higher-tier law.

Tiering should constrain extension rather than confer trust by label. Each operation schema must bind input types, required authority, evidence, decision outcomes, effect class, receipt fields, failure semantics, version, and compatibility rules. An application-defined operation cannot revise kernel law merely by registering a name.

8. Receipts and the Three-Hash Interface

Receipts form a governance history only when they bind exact objects and transitions. A useful receipt distinguishes at least:

- request identity and content;
- governing charter and policy version;
- authority and delegation evidence;
- decision and reason;
- pre-effect commitment;
- observed execution result;
- predecessor and revision lineage;
- verifier and canonicalization rules.

The Three Hash Rule supplies a relevant integrity interface: identity, state, and evidence or transition commitments should not collapse into one undifferentiated digest. Hashes can support exact binding, tamper evidence, and replay comparison. They do not prove truth, authority, semantic equivalence, or legitimacy without the surrounding interpretation and verification rules.

9. Authority Laundering

Authority laundering occurs when a result gains perceived legitimacy through workflow position, repetition, a trusted-looking container, successful execution, or vague upstream approval without a reconstructable authority-granting event.

UGK resists laundering by preserving distinct links:

```
actor -> represented principal -> intent -> mandate -> authority
      -> policy -> evidence -> decision -> commitment -> effect -> receipt
```

The test is not merely whether this chain exists in data. The system should ask whether the effect remains reachable if the asserted authority or delegation is removed. If it does, the claimed authority may be decorative rather than causal.

10. Core Invariants

1. **No effect before decision commitment.**
2. **Refusal implies no admitted effect.**
3. **Every terminal disposition is receipted, including refusal and unknown.**
4. **Identity, capability, will, authority, decision, execution, and receipt remain distinct.**
5. **Authority is explicitly bounded by action, target, scope, time, jurisdiction, delegation, and revocation.**
6. **Execution cannot widen the committed effect.**
7. **Application schemas cannot silently amend kernel law.**
8. **Receipts bind exact versions and canonicalization rules.**
9. **Successful execution does not retroactively create authority.**
10. **Cryptographic integrity does not establish factual truth or moral legitimacy.**
11. **Recovery and amendment preserve prior history.**
12. **Unknown observation remains unknown rather than being rewritten as success or failure.**

11. Relation to the Corpus

Authority Laundering Theory supplies the adversarial diagnosis. *Reasoning by Way of Receipts* constrains conclusions to retained evidence. *The Three Hash Rule* supplies a compact integrity discipline. *Constitutional Computing Architecture* places UGK within a larger reorganization of computing around inspectable and renegotiable authority.

The Governed Meaning Base can represent identities, authorities, charters, schemas, decisions, receipts, conflicts, and revision lineage. It does not become the Governor merely by storing them. These are typed interfaces, not identity mergers or proof of a particular implementation.

12. Realization Funnel

$$
\begin{aligned}
&\mathit{Possibility}\\
&\to \mathit{Specification}\\
&\to CM_1\\
&\to CM_2\\
&\to IC\\
&\to IR\\
&\to \mathit{Evidence}\\
&\to \mathit{Revision}
\end{aligned}
\quad .
$$

- **Possibility:** enumerate governance regimes, authority models, operation types, beneficial uses, captured systems, and harmful realizations.
- **Specification:** define distinctions, roles, event semantics, state transitions, schemas, invariants, receipts, and failure outcomes.
- **CM1:** retain all admissible configurations, including centralized, ceremonial, opaque, coercive, insecure, and administratively excessive forms.
- **CM2:** select preferable forms under legitimacy, least authority, safety, inspectability, performance, privacy, portability, refusal, recovery, and user sovereignty criteria.
- **IC:** bind an Instance Constitution or Instance Configuration for a domain, charter, identities, policies, schema versions, and effect boundary.
- **IR:** construct its Instance Realization or Implementation Realization as a particular kernel and governed application environment.
- **Evidence:** verify fixtures, courts or test suites, receipt chains, effect boundaries, refusals, recovery, and adversarial cases.
- **Revision:** amend charter, schemas, authority, implementation, and claims through retained lineage and explicit authorization.

This educational brief supplies a candidate Specification and CM1/CM2 vocabulary. It does not select an IC, identify one canonical IR, or verify a deployed realization. Realization-stage IR remains distinct from Intermediate Representation in compiler contexts.

Accountable Authority Transaction Kernel

Across the completed P1 corpus, the Universal Governance Kernel is the **accountable-authority transaction kernel for converting a proposed intent into a bounded commitment, attempted effect, and claim-limited receipt under a declared charter**. Its universal element is the distinction

grammar and transaction pattern, not a claim that one kernel, authority model, or implementation should govern every domain.

- **Adjacent work:** *Authority Laundering Theory*
  Supplies: Diagnosis of authority gained through workflow position, trusted containers, repetition, or successful effect.
  UGK boundary: UGK supplies a transaction structure in which authority can be tested rather than presumed.
- **Adjacent work:** *Reasoning by Way of Receipts*
  Supplies: Claim-bounded inference from proposals, decisions, effects, and evidence.
  UGK boundary: UGK emits governance events; the receipt theory governs what may be inferred from them.
- **Adjacent work:** *Jnana II: Constitutional Transformation Substrate*
  Supplies: A software transformation substrate with capabilities and effect membranes.
  UGK boundary: A Jnana profile may host UGK functions; neither identity entails the other.
- **Adjacent work:** *The Mechanics of Constitutional Enforcement*
  Supplies: Complete mediation, effect atomicity, classified remainder, and repair.
  UGK boundary: UGK defines the transaction; enforcement determines whether effects can bypass it.
- **Adjacent work:** *Governance Coverage Gaps*
  Supplies: Capability and grant-path coverage diagnosis.
  UGK boundary: UGK conformance claims are limited to paths shown to cross the kernel.

The kernel transaction should bind:

$$K = (charter, principal, intent, will, authority, evidence,\\ decision, commitment, executor, effect, observation,.\\ receipt, contest, revision)$$

The pre-effect commitment is the constitutional hinge. It must fix the admitted request, authority, exact target, scope, duration, capabilities, conditions, expected effect class, and failure semantics before the effect occurs. Execution may return `EFFECT_CONFIRMED`,
`EFFECT_FAILED`, `EFFECT_PARTIAL`,
or `EFFECT_UNKNOWN`; none changes the earlier authority retroactively.

Logical Steward, Governor, Executor, and Witness functions may share a process only if role-specific inputs, authorities, decisions, and evidence remain distinguishable and independently challengeable where required. A charter is itself an

object of authority, amendment, appeal, rights, and legitimacy review. Founding cannot self-certify its moral, legal, or institutional validity.

The universal GSM can type
`PROPOSES_UNDER_STEWARD_ROLE`, `REPRESENTS_PRINCIPAL_WILL`,
`EVALUATES_UNDER_GOVERNOR_ROLE`, `COMMITS_BEFORE_EFFECT`, `EXECUTES_WITHIN_COMMITMENT`,
`OBSERVES_PARTIAL_EFFECT`, `CHALLENGES_KERNEL_DECISION`, `AMENDS_CHARTER`,
`BYPASSES_GOVERNANCE_KERNEL`,
and
`REMAINS_IDENTITY_UNADJUDICATED_AMONG_UGK_`
`REALIZATIONS`. These relations preserve the architecture while respecting the existing identity hold.

This integration establishes four editorial commitments: retain UGK as the accountable-authority transaction kernel; adopt the expanded transaction object and four effect outcomes; require logical role and contest separation even when components are colocated; and preserve all UGK system, profile, codex, and implementation identities as unmerged until separately adjudicated.

Transaction, Representation, and Orientation State

The fixed-seed comparison with *MCIR Topologies* and *Epistemic Situational Awareness and the Governed Knowledge Loop* exposes three states that a UGK deployment must not collapse:

$$G_t = (transaction_state, representation_state, epistemic_orientation_t).$$

The transaction state records the request, authority, decision, commitment, attempted effect, observation, and receipt. The representation state records which topology, schema, canonicalization, projection, and interpretation rules give those objects their machine-operable form. The epistemic orientation records which evidence, conflicts, unknowns, authority facts, and environmental observations are current enough for the proposed continuation.

MCIR topology is therefore upstream of some UGK integrity claims without becoming the Governor. If a projection aliases two authorities, drops a revocation edge, hides a conflict, or silently changes shared-node ownership, a structurally valid kernel transaction can operate on the wrong represented constitution. Conversely, an MCIR artifact cannot authorize an effect merely because it represents one. UGK must bind the exact representation regime used at decision and execution time and receipt any topology migration that can alter interpretation.

ESA informs whether the transaction is sufficiently located to proceed, but it does not bind the decision or supply authority. A current orientation can support
`ADMIT`,
`DEFER`, `REFUSE`, `CONFLICT`,

or EFFECT_UNKNOWN; the Governor remains responsible for the disposition under the charter. Stale or incomplete orientation is a reason to qualify or stop, not a hidden substitute for judgment.

These are optional architectural interfaces, not mandatory product dependencies. A minimal UGK can use other representation and orientation mechanisms if it preserves the same distinctions and evidence. Two additional failure tests follow: remove or alter a topology edge and verify that authority-sensitive interpretation cannot change silently; stale one required ESA dimension and verify that continuation cannot retain an unqualified admission.

The universal GSM can type
BINDS_TRANSACTION_TO_REPRESENTATION_REGIME,
PROJECTS_AUTHORITY_WITH_REMAINDER, ORIENTS_GOVERNANCE_DECISION,
DETECTS_STALE_DECISION_DIMENSION, MIGRATES_TOPOLOGY_UNDER_COMMITMENT,
and
DOES_NOT_DERIVE_AUTHORITY_FROM_REPRESENTATION_
OR_ORIENTATION.

This randomized review adds three open editorial questions: adopt the three-state separation; require exact representation-regime custody for authority-sensitive transactions; and add topology-mutation and stale-orientation tests without making MCIR or ESA mandatory UGK implementations.

Kernel Contract, Profiles, and Realizations

The frozen P3 field uses UGK as a reusable transaction grammar, an optional implementation family, a governance substrate inside a larger observability platform, an infrastructure scheduling example, an enforcement host, an operating-system role, a development-process component, and a domain profile for international trade. These are different relations and standings.

The stable kernel contract is:

$$\begin{aligned} K = (&charter, subject, intent, purpose, authority, evidence,\\ &disposition, commitment, effect_attempt, observation,.\\ &receipt, contest, successor, remainder) \end{aligned}$$

A profile specializes this contract for a domain, consequence class, authority topology, privacy regime, schema family, effect boundary, and remedy. A realization selects components, code, deployment boundaries, witnesses, and tests. Neither a profile name nor a working component proves kernel coverage, conformance, legitimacy, or non-bypass.

AFS can express a UGK profile without proving an implementation. Agentic Interface Surfaces can project selected kernel constraints and outcomes without becoming the Governor. COP can use UGK as a governance substrate while retaining independent evidence-health and coverage roles. Constitutionally

Governed Infrastructure can use UGK scheduling as one bounded case while keeping liveness, access, remedy, and affected-party evidence outside the kernel receipt.

ACG, Jnana, the enforcement mechanics, MosaiCOS, and IEL each provide a possible host or surrounding lifecycle. Their downstream uses require explicit conformance versions and path coverage rather than identity merger. Legitimacy Without Self-Ratification limits any kernel's ability to validate its own authority or evidence. APD limits effect-containment claims to the paths actually mediated. The international-trade profile specializes the grammar while remaining on title and identity hold.

The universal GSM can type
`IMPLEMENTS_UGK_CONTRACT`, `SPECIALIZES_UGK_PROFILE`,
`HOSTS_KERNEL_FUNCTION`, `PROJECTS_KERNEL_INTERFACE`, `SURROUNDS_WITH_EVIDENCE_LOOP`,
`TESTS_KERNEL_PATH_COVERAGE`, `WITHHOLDS_SELF_RATIFICATION`,
and
`RETAINS_UGK_IDENTITY_HOLD`.

This reverse-dependency review establishes four editorial commitments: define the minimal UGK contract independently of any named realization; adopt a typed profile and conformance relation; require path, version, consequence, and evidence coverage for every realization claim; and keep all UGK systems, codices, profiles, and implementations unmerged until separately adjudicated.

13. Counterevidence and Failure Tests

The architecture must narrow, revise, or fail if: (1) a permission check already establishes every claimed distinction; (2) will is used to manufacture authority; (3) identity is treated as authority; (4) capability is treated as permission; (5) execution occurs before commitment; (6) refusal still permits the effect; (7) admission is recorded only after the effect; (8) result evidence is absent; (9) an effect failure hides partial side effects; (10) unknown observation is reported as success; (11) the Governor can widen its own authority; (12) the Steward approves its own request without an admitted rule; (13) the Executor widens target or scope; (14) combined roles erase logical separation; (15) a charter self-certifies legitimacy; (16) founding provenance is missing; (17) delegation has no scope or expiry; (18) revocation does not block later use; (19) alternate authority paths bypass the declared Governor; (20) a schema tier changes kernel law; (21) version incompatibility is ignored; (22) canonicalization is ambiguous; (23) hashes verify altered interpretations; (24) a receipt is mistaken for truth; (25) a receipt is mistaken for moral legitimacy; (26) a successful effect launders authority; (27) refusal receipts disclose protected evidence; (28) privacy rules make audit impossible; (29) audit requirements make ordinary use impractical; (30) recovery rewrites history; (31) appeals cannot challenge a decision; (32) the authority-removal test leaves the effect reachable; (33) the architecture cannot represent conflict among authorities; (34) one system candidate is silently

declared the canonical UGK; (35) an implementation claims conformance without a versioned contract and tests; (36) cryptographic claims are made without executable verification; (37) external prior art contradicts novelty claims; (38) human legitimacy questions are delegated to the kernel; (39) GMB storage is confused with governing authority; or (40) publication standing is inferred from architectural coherence.

A Governance Kernel Can Expose Authority Claims Without Making Them Legitimate

Public orientation

UGK separates intent, will, authority, decision, commitment, execution, and result receipt. Steward, Governor, and Executor are functions; a charter and schema registry bound ordinary effects. The architecture can make a claimed authority path inspectable. It cannot manufacture legitimate purpose, deserved authority, true evidence, wise decisions, or correct effects from well-formed records.

Adversarial tests

1. **Charter-legitimacy attack.** A declared or hashed charter is treated as lawful, consensual, ethical, representative, or properly founded.
2. **Role-collusion attack.** Steward, Governor, and Executor are nominally separate but share control, incentives, credentials, infrastructure, or failure modes.
3. **Purpose-laundering attack.** A request is reframed into an authorized purpose while the affected subject's will or jurisdiction is absent.
4. **Schema-capture attack.** Registry owners determine which operations, evidence, refusals, and harms can be represented.
5. **Decision-effect gap attack.** Execution differs from the committed decision, or effects occur through an unmediated path.
6. **Receipt-proof attack.** A transaction envelope establishes only recorded lineage but is promoted into proof of legitimacy, correctness, or real-world outcome.
7. **Emergency-power attack.** Overrides, degradation, or break-glass authority lacks expiry, review, repair, and successor discipline.
8. **Universal-kernel attack.** One governance grammar is imposed across domains whose authority, rights, evidence, remedies, and institutional forms differ.

Claim and evidence ceiling

The paper specifies an accountable-authority transaction kernel and failure tests. It does not establish legitimate founding, complete role separation, correct decisions, complete mediation, actual effects, cryptographic security, legal validity,

universal applicability, or a conforming implementation. Multiple UGK-named systems and profiles remain distinct until author adjudication.

Every domain profile requires its own authority source, rights and nondelegable protections, evidence rules, remedies, exceptions, affected-party standing, bypass tests, and implementation evidence. No conformance review, deployment, identity merger, or external research occurred in this pass.

Reader-facing consequence

UGK should let a person ask whose purpose an action serves, who could bind the decision, what evidence was admitted, what was refused, what effect actually occurred, and how to challenge it - while preserving the possibility that the entire authority claim is invalid.

Conclusion

UGK reframes execution as a constitutional event. An action is not adequately governed merely because a capable actor passed a permission check. Its intent, represented purpose, authority, evidence, decision, commitment, effect, and receipt should remain separately inspectable.

That discipline can make responsibility harder to hide. It cannot make authority legitimate by declaration or ethics automatic by computation. The kernel's proper role is to preserve the structure through which legitimacy claims can be tested, challenged, revised, and refused before effects become irreversible.

Reference and Verification Note

Author corpus interfaces inspected under exact custody

Ableman Adam Ableman *Authority Laundering Theory.*

Ableman Adam Ableman *Reasoning by Way of Receipts.*

Ableman Adam Ableman *The Three Hash Rule.*

Ableman Adam Ableman *Constitutional Computing Architecture.*

External verification required

No external literature was consulted. Publication comparison should cover access-control and authorization models, policy engines, workflow systems, reference monitors, capability security, event sourcing, audit logging, tamper-evident logs, distributed ledgers, constitutional and administrative governance, delegation, revocation, and accountable execution. Novelty, security, legality, and conformance remain unestablished.

Realization Scope and Evidence Boundary

This successor makes the paper's stage boundary explicit. It narrows claims rather than manufacturing missing evidence: a stage named as context is not thereby claimed as implemented, validated, or complete.

- **Revision.** Existing source passages use this stage only as bounded context or an obligation. The paper does not claim an independently validated contribution at this stage.
- **Publication form.** This work is retained as a standalone technical companion because its concrete migration method is useful independently; it remains subordinate to the architecture it explains.

Corpus-Wide Reference Positioning

This paper argues that the Universal Governance Kernel (UGK) is a governance-first execution architecture. Its nearest external comparison fields are governance, security, provenance. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: A written constitution can legitimate domination as easily as constrain it unless amendment, participation, enforcement, and appeal remain effective in practice. Unforgeable capabilities can constrain exercise of authority while still leaving allocation, delegation, revocation, and social legitimacy incorrectly specified. A complete provenance graph can faithfully record a false or unauthorized process; provenance is evidence about lineage, not automatic truth or legitimacy. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] Elinor Ostrom. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press, 1990. DOI: 10.1017/CBO9780511807763. https://doi.org/10.1017/CBO9780511807763 **Relationship:** Predecessor institutional account of rule-governed collective action and polycentric governance.

[**E2**] Wil Waluchow and Dimitrios Kyritsis. "Constitutionalism." *Stanford Encyclopedia of Philosophy*, substantive revision 2023. https://plato.stanford.edu/entries/constitutionalism/ **Relationship:** Adjacent legal-philosophical account of limited authority; included to expose differences between political and computational constitutionalism.

[**E3**] Lawrence Lessig. *Code: Version 2.0*. Basic Books, 2006. https://codev2.cc/ **Relationship:** Predecessor argument that software architecture regulates behavior; the paper's contribution may be to make such regulation inspectable and revisable.

[**E4**] OECD. *Recommendation of the Council on Artificial Intelligence*. OECD/LEGAL/0449, adopted 2019 and updated 2024. legalinstruments.oecd.org source **Relationship:** Adjacent policy standard for accountability, traceability, transparency, safety, and human-centered values.

[**E5**] Jerome H. Saltzer and Michael D. Schroeder. "The Protection of Information in Computer Systems." *Proceedings of the IEEE* 63(9), 1975, pp. 1278-1308. DOI: 10.1109/PROC.1975.9939. https://doi.org/10.1109/PROC.1975.9939 **Relationship:** Foundational protection principles used as a comparison point for least authority, mediation, and inspectability.

[**E6**] Mark S. Miller. *Robust Composition: Towards a Unified Approach to Access Control and Concurrency Control*. PhD dissertation, Johns Hopkins University, 2006. http://www.erights.org/talks/thesis/ **Relationship:** Predecessor object-capability treatment of authority and compositional security.

[**E7**] Gerwin Klein et al. "seL4: Formal Verification of an OS Kernel." *SOSP '09*, 2009, pp. 207-220. DOI: 10.1145/1629575.1629596. https://doi.org/10.1145/1629575.1629596 **Relationship:** Predecessor for machine-checked implementation correspondence and capability-based kernel design.

Ableman corpus

[**A1**] Adam Ableman. *Ableman Constitutional Governance*. Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-79353AF163A4B0F1;
source SHA-256
FB0D9B2D3D54168FFE70B24308D1B2766D6D663ED24502D475E975939BE97C8B.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Constitutional Theory*. Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-D2D97C4042FD281F;
source SHA-256
FACC256EDDE5A7D7EA18133CFC9C060CFB8C3811B0221E1969E62347F2BB69CF.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Governance Engineering*. Ableman Corpus current edition; work identity PAPER.PUB.18749870;
source SHA-256
372169F315D192733856B1F4A3F1038BBC73F57CF3ED9ACABAE72B965A69C8D0.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Legitimacy Without Self-Ratification*. Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-EF2382F9397335AC;
source SHA-256
0186A39C808F8FDDB5C8F42384DF1A26678928434BD50A0EAE8D4947390F68A2.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Effect Sovereignty*. Ableman Corpus current edition; work identity PAPER.PUB.18737323;
source SHA-256
1DB8A34946A8705F6F4BA4FC3B12B251FC7C83DBDF217F9DDFF55529C8612EE7.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *Receipt-Bound Capabilities: Membrane Tokens for Governed Interaction with Reality*. Ableman Corpus current edition; work identity PAPER.PUB.18601631;
source SHA-256
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Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

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Editorial source notes retained from predecessor

The entries below are corpus-internal comparison and dependency sources. Their inclusion does not establish external validation of this paper's empirical, safety, novelty, or performance claims.

Adam Ableman *The Three Hash Rule.* Zenodo, 2026. doi:10.5281/zenodo.20788617

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Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus
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